CHAPTER 2

# THE ORIGIN AND EVOLUTION OF THE EARTH

**CHAPTER ESSENCE AT A GLANCE:**
Explore the origin of the universe through the Big Bang theory, planetary formation from the solar nebula, density differentiation creating Earth's layered interior, and the evolution of the atmosphere, oceans, and life.

## 4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Early Hypotheses and Nebular Theory:** Immanuel Kant and Pierre-Simon Laplace proposed that planets condensed out of a slowly rotating cloud of gas (nebula); later modified by Carl Weizsacker and Otto Schmidt acknowledging friction and accretion.

• **The Big Bang Model of Cosmic Expansion:** Georges Lemaitre and Edwin Hubble (1929) established that the universe originated 13.7 billion years ago from an infinitely dense singularity, expanding continuously and cooling to form atoms, stars, and galaxies.

• **Planetary Formation via Accretion:** Within nebular gas clouds, gravitational instabilities produced rotating gaseous discs; particles condensed into small rounded bodies (planetesimals) which accreted through collision into planets 4.6 billion years ago.

• **Evolution of Lithosphere, Atmosphere and Oceans:** Density differentiation caused heavy metals (iron and nickel) to sink into the core while lighter silicates floated; volcanic degassing released water vapour and carbon dioxide, which condensed into primordial oceans 3.8 billion years ago.

## ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$$\text{Hubble Expansion Law: } v = H_0 \cdot d \rightarrow \text{Recession Velocity} \propto \text{Distance}$$

*Cosmological Evidence of Expanding Galactic Space*

$$F_g = G\frac{m_1 m_2}{r^2} \rightarrow \text{Gravitational Accretion and Clumping}$$

*Newtonian Accretion Force Governing Planetesimal Growth*

$$\text{Atmospheric Evolution: Primordial} \rightarrow \text{Volcanic Degassing} \rightarrow \text{Photosynthesis (Free } O_2\text{)}$$

*Three-Stage Genesis of Terrestrial Atmosphere*

$$\text{Planetary Density Contrast: Terrestrial (High } \rho) \longleftrightarrow \text{Jovian (Low } \rho)$$

*Inner Terrestrial and Outer Jovian Structural Divide*

## CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Earth's primordial atmosphere contained negligible free oxygen; oxygen accumulated in large amounts only after photosynthetic blue-green algae (cyanobacteria) proliferated around 2.5 to 3.0 billion years ago.

• **Alert:** Terrestrial planets (Mercury, Venus, Earth, Mars) are rocky because intense solar wind blew away lighter gases close to the Sun, whereas Jovian gas giants formed at distant, cooler orbits.

• **Alert:** Density differentiation is the density-driven gravitational separation of molten Earth into core, mantle, and crust, occurring around 4.6 billion years ago.

## HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

### 1. What is the Big Bang Theory? Who provided observational evidence for the expanding universe? [Score: 2 Marks]

**Answer & Key Valuation Steps:**

• **Big Bang Theory:** Also called the expanding universe hypothesis; states that all matter in the universe was concentrated in one tiny point ('singular atom') of infinite temperature and density that exploded 13.7 billion years ago. **[1 Mark]**

• **Observational Evidence:** Edwin Hubble provided observational evidence in 1929 by proving that galaxies are moving away from each other with velocities proportional to distance. **[1 Mark]**

### 2. Why are the inner planets called Terrestrial and the outer planets called Jovian? [Score: 2 Marks]

**Answer & Key Valuation Steps:**

• **Terrestrial Planets:** Mercury, Venus, Earth, and Mars are called terrestrial because they are rocky, dense, and possess solid crusts similar to Earth. **[1 Mark]**

• **Jovian Planets:** Jupiter, Saturn, Uranus, and Neptune are called Jovian (Jupiter-like) because they are giant gaseous bodies with thick atmospheres of hydrogen and helium and low densities. **[1 Mark]**

### 3. Explain the Nebular Hypothesis regarding the origin of the Earth proposed by Laplace. [Score: 3 Marks]

**Answer & Key Valuation Steps:**

• **Core Proposition:** Pierre-Simon Laplace revised Kant's idea in 1796, hypothesizing that planets were formed out of a slowly rotating cloud of gaseous matter (nebula) associated with a youthful Sun. **[1 Mark]**

• **Cooling and Rings:** As the nebula cooled, it contracted and spun faster, throwing off concentric rings of gaseous material due to centrifugal force. **[1 Mark]**

• **Condensation into Planets:** The material within each ejected ring condensed gravitationally to form individual planets, while the central core remained as the Sun. **[1 Mark]**

### 4. Describe the three stages in the evolution of the Earth's present atmosphere. [Score: 3 Marks]

**Answer & Key Valuation Steps:**

• **Stage 1 (Loss of Primordial Atmosphere):** Intense solar winds from the young Sun stripped away the primordial atmosphere of hydrogen and helium from inner planets. **[1 Mark]**

• **Stage 2 (Degassing):** Massive internal heating caused volcanic eruptions that released water vapour, nitrogen, carbon dioxide, methane, and ammonia, but very little free oxygen. **[1 Mark]**

• **Stage 3 (Photosynthetic Modification):** Water vapour condensed into oceans; primitive life evolved around 3.8 billion years ago, and photosynthesis saturated oceans and atmosphere with free oxygen. **[1 Mark]**

### 5. Explain the process of differentiation that led to the development of the layered internal structure of the Earth. [Score: 4 Marks]

**Answer & Key Valuation Steps:**

• **Initial State of Earth:** The Earth was mostly volatile, rocky, and hot during its primordial formative accretion period 4.6 billion years ago. **[1 Mark]**

• **Temperature Surge:** Due to gravitational compression, radioactive decay, and meteorite impacts, internal temperature increased drastically, partially melting the interior. **[1 Mark]**

• **Density Separation:** Dense materials like metallic iron and nickel sank towards the Earth's center, while lighter siliceous materials floated towards the surface. **[1 Mark]**

• **Concentric Layering:** With subsequent cooling, this differentiation created concentric layers from surface to center: crust, mantle, outer core, and inner core. **[1 Mark]**

**6. Describe the successive stages involved in the formation of planets from a rotating nebular cloud. [Score: 4 Marks]**

**Answer & Key Valuation Steps:**

• **Stage 1 (Localized Gas Clumps):** Stars are localized lumps of gas within a nebula; gravitational force within the lumps leads to the formation of a dense rotating gas core and surrounding dust disc. **[1 Mark]**

• **Stage 2 (Condensation of Planetesimals):** The gas cloud starts condensing, and matter around the core develops into small rounded objects through cohesion called planetesimals. **[1.5 Marks]**

• **Stage 3 (Gravitational Accretion):** Large numbers of small planetesimals collide and coalesce under mutual gravitational attraction to form a smaller number of large planetary bodies. **[1.5 Marks]**